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PROFESSIONAL SUMMARY

Driven Chemical Engineering graduate and Materials Researcher with hands-on experience in thin-film deposition, electrochemical engineering, and AI-assisted research platforms. Experienced in bridging experimental materials characterization with Python-driven data analysis and computational workflows. Seeking Master's programs in Materials, Energy, or Chemical Engineering at German universities (Summer 2027).

WORK EXPERIENCE

N-ERGY AI SOLUTIONS INDIA PVT LTD

Materials associate | Oct 2025 - Present

- Building automated literature ingestion pipeline using OpenAlex API to search, retrieve, and index open-access scientific papers into a vector database for the MatGeek RAG agent
- Developing interactive data visualization interfaces using Plotly to display simulation and experimental results within the platform
- Developed and deployed Python UQ classes to standardize automated analysis of complex materials datasets.

Experimental Materials associate | Feb 2025 - Oct 2025

- Optimized Aluminum thin-film deposition using RF sputtering to achieve uniform 100 nm films for micro-device architectures, controlling room-temperature processing parameters.
- Experienced in materials synthesis through sol-gel and metallurgical routes, with hands-on expertise in experimental testing for advanced material applications.

RESEARCH EXPERIENCE

Undergraduate Researcher – Funded Project | SSN College, Chennai, India | Jan 2024 – Oct 2024

Guide: Dr. A.K.Lakshminarayanan

- Investigated the electrochemical kinetics and pitting corrosion behavior of WAAM 316L stainless steel using advanced potentiodynamic polarization and microscopy
- Project funded with ₹35,000, supporting experimental design and material characterization.

Project Intern | BRAKES INDIA PVT LTD, Chennai, India | Oct 2024 – Feb 2025

Guide: Dr. Pachimatla Rajesh

- Acquired hands-on expertise in surface finishing, defect analysis, and process optimization within industrial electrochemical treatments.
- Optimized industrial electrochemical anodizing processes for automotive components, improving surface microfinish by 78% and mitigating mass transport/diffusion defects within the electrolyte bath.

PUBLICATIONS/CONFERENCES:

Presented research at the National Conference on Innovation of Process Engineering for its Sustainable Development, IChE-CRC, March 2025. Paper: "*Enhancing Microfinish and Resolving Smut and Powdery Coating Issues in the Anodizing Process for Automotive Plunger Manufacturing*"

ACADEMIC HISTORY

SSN College of Engineering | 2021-2025

Bachelor of Technology in Chemical Engineering

- CGPA: 8.3/10 (equivalent to ~3.3/4 GPA)
- Department 3rd Rank

SBOA School and Junior College | 2019-2021

High school

- Percentage: 88%

ACADEMIC STRENGTHS

- Transport Phenomena (Fluid Mechanics & Mass Transfer), Electrochemical Engineering, Process Dynamics and Control, Chemical Reaction Engineering.

TECHNICAL SKILLS

- Characterization: FTIR, XPS, SEM/EDS, XRD, AFM, advanced expertise in structural, chemical, and surface analysis of functional materials.
- Fabrication & Synthesis: High-entropy alloys (HEA), colloidal synthesis, solid-state metallurgy, additive arc melting, RF-Sputtering, CO₂ laser processing.
- Data Analysis & Modeling: Python (Pandas, Plotly), MATLAB, Streamlit, OriginLab, Aspen Plus.
- Developer Tools: Git/GitHub, PyMongo, UI/UX (Streamlit Sandbox deployment).